Netflix_Analysis_By_Kumud

December 10, 2024

[1]:
```
import pandas as pd
df = pd.read_csv(r"C:\Users\kpswe\OneDrive\Desktop\netflix.csv")
df.head(5)
```

[1]:
```
  show_id     type                   title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                cast        country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…            NaN
3                                                NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

           date_added  release_year rating   duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                           description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[2]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

[3]:
```
df.isna().sum()
```

[3]:
```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

[4]:
```
df.isna().sum(axis=1).sum()
```

[4]: 4307

[5]:
```
df.isnull().any(axis=1).sum()
```

[5]: 3475

[6]:
```
df[df['duration'].isna()]
```

[6]:
```
     show_id   type                                title    director  \
5541   s5542  Movie                       Louis C.K. 2017  Louis C.K.
5794   s5795  Movie                  Louis C.K.: Hilarious  Louis C.K.
5813   s5814  Movie  Louis C.K.: Live at the Comedy Store  Louis C.K.

            cast        country          date_added  release_year  rating  \
5541  Louis C.K.  United States       April 4, 2017          2017  74 min
5794  Louis C.K.  United States  September 16, 2016          2010  84 min
5813  Louis C.K.  United States     August 15, 2016          2015  66 min

     duration listed_in                                       description
5541      NaN    Movies  Louis C.K. muses on religion, eternal love, gi…
5794      NaN    Movies  Emmy-winning comedy writer Louis C.K. brings h…
5813      NaN    Movies  The comic puts his trademark hilarious/thought…
```

[7]:
```
df.loc[df['show_id'].isin(['s5542', 's5795', 's5814']), 'duration'] =␣
↪df['rating']
df[df['show_id'].isin(['s5542','s5795','s5814'])]
```

[7]:
```
     show_id   type                                title    director  \
5541   s5542  Movie                       Louis C.K. 2017  Louis C.K.
5794   s5795  Movie                  Louis C.K.: Hilarious  Louis C.K.
5813   s5814  Movie  Louis C.K.: Live at the Comedy Store  Louis C.K.

            cast        country          date_added  release_year  rating  \
5541  Louis C.K.  United States       April 4, 2017          2017  74 min
5794  Louis C.K.  United States  September 16, 2016          2010  84 min
5813  Louis C.K.  United States     August 15, 2016          2015  66 min

     duration listed_in                                       description
5541   74 min    Movies  Louis C.K. muses on religion, eternal love, gi…
5794   84 min    Movies  Emmy-winning comedy writer Louis C.K. brings h…
5813   66 min    Movies  The comic puts his trademark hilarious/thought…
```

[8]:
```
import numpy as np
df.loc[df['show_id'].isin(['s5542', 's5795', 's5814']), 'rating']= np.nan
df[df['show_id'].isin(['s5542','s5795','s5814'])]
```

[8]:
```
     show_id   type                                title    director  \
5541   s5542  Movie                       Louis C.K. 2017  Louis C.K.
5794   s5795  Movie                  Louis C.K.: Hilarious  Louis C.K.
5813   s5814  Movie  Louis C.K.: Live at the Comedy Store  Louis C.K.

            cast        country          date_added  release_year rating  \
5541  Louis C.K.  United States       April 4, 2017          2017    NaN
5794  Louis C.K.  United States  September 16, 2016          2010    NaN
5813  Louis C.K.  United States     August 15, 2016          2015    NaN
```

```
     duration listed_in                                        description
5541   74 min    Movies  Louis C.K. muses on religion, eternal love, gi…
5794   84 min    Movies  Emmy-winning comedy writer Louis C.K. brings h…
5813   66 min    Movies  The comic puts his trademark hilarious/thought…
```

[9]:
```
df[df['date_added'].isna()]
```

[9]:
```
     show_id     type                                          title director  \
6066   s6067  TV Show  A Young Doctor's Notebook and Other Stories      NaN
6174   s6175  TV Show              Anthony Bourdain: Parts Unknown      NaN
6795   s6796  TV Show                                       Frasier      NaN
6806   s6807  TV Show                                       Friends      NaN
6901   s6902  TV Show                                Gunslinger Girl      NaN
7196   s7197  TV Show                                       Kikoriki      NaN
7254   s7255  TV Show                           La Familia P. Luche      NaN
7406   s7407  TV Show                                          Maron      NaN
7847   s7848  TV Show                                   Red vs. Blue      NaN
8182   s8183  TV Show                  The Adventures of Figaro Pho      NaN

                                                   cast         country  \
6066  Daniel Radcliffe, Jon Hamm, Adam Godley, Chris…  United Kingdom
6174                                   Anthony Bourdain   United States
6795  Kelsey Grammer, Jane Leeves, David Hyde Pierce…   United States
6806  Jennifer Aniston, Courteney Cox, Lisa Kudrow, …   United States
6901  Yuuka Nanri, Kanako Mitsuhashi, Eri Sendai, Am…           Japan
7196                                     Igor Dmitriev             NaN
7254  Eugenio Derbez, Consuelo Duval, Luis Manuel Áv…   United States
7406  Marc Maron, Judd Hirsch, Josh Brener, Nora Zeh…   United States
7847  Burnie Burns, Jason Saldaña, Gustavo Sorola, G…   United States
8182  Luke Jurevicius, Craig Behenna, Charlotte Haml…       Australia

     date_added  release_year rating    duration  \
6066        NaN          2013  TV-MA   2 Seasons
6174        NaN          2018  TV-PG   5 Seasons
6795        NaN          2003  TV-PG  11 Seasons
6806        NaN          2003  TV-14  10 Seasons
6901        NaN          2008  TV-14   2 Seasons
7196        NaN          2010   TV-Y   2 Seasons
7254        NaN          2012  TV-14   3 Seasons
7406        NaN          2016  TV-MA   4 Seasons
7847        NaN          2015     NR  13 Seasons
8182        NaN          2015  TV-Y7   2 Seasons

                                            listed_in  \
6066           British TV Shows, TV Comedies, TV Dramas
6174                                       Docuseries
```

```
6795                     Classic & Cult TV, TV Comedies
6806                     Classic & Cult TV, TV Comedies
6901                       Anime Series, Crime TV Shows
7196                                           Kids' TV
7254  International TV Shows, Spanish-Language TV Sh…
7406                                       TV Comedies
7847  TV Action & Adventure, TV Comedies, TV Sci-Fi …
8182                              Kids' TV, TV Comedies

                                           description
6066  Set during the Russian Revolution, this comic …
6174  This CNN original series has chef Anthony Bour…
6795  Frasier Crane is a snooty but lovable Seattle …
6806  This hit sitcom follows the merry misadventure…
6901  On the surface, the Social Welfare Agency appe…
7196  A wacky rabbit and his gang of animal pals hav…
7254  This irreverent sitcom featues Ludovico, Feder…
7406  Marc Maron stars as Marc Maron, who interviews…
7847  This parody of first-person shooter games, mil…
8182  Imagine your worst fears, then multiply them: …
```

[10]:
```
df = df.dropna(subset=['date_added'])
```

[11]:
```
df.isna().sum()
```

[11]:
```
show_id            0
type               0
title              0
director        2624
cast             825
country          830
date_added         0
release_year       0
rating             7
duration           0
listed_in          0
description        0
dtype: int64
```

[12]:
```
import re

invalid_dates = df[~df['date_added'].str.match(r"^[A-Za-z]+ \d{1,2}, \d{4}$")]
invalid_dates['date_added']
```

[12]:
```
6079        August 4, 2017
6177     December 23, 2018
6213     December 15, 2018
```

```
6279          July 1, 2017
6304         July 26, 2019
               …
8539       January 8, 2018
8557     December 25, 2015
8684        August 1, 2017
8712        March 31, 2017
8755      November 1, 2016
Name: date_added, Length: 88, dtype: object
```

[13]:
```python
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed')
df.head(5)
```

[13]:
```
  show_id     type                  title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                cast        country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…            NaN
3                                                NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                        description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[14]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8797 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8797 non-null   object
 1   type          8797 non-null   object
 2   title         8797 non-null   object
 3   director      6173 non-null   object
 4   cast          7972 non-null   object
 5   country       7967 non-null   object
 6   date_added    8797 non-null   datetime64[ns]
 7   release_year  8797 non-null   int64
 8   rating        8790 non-null   object
 9   duration      8797 non-null   object
 10  listed_in     8797 non-null   object
 11  description   8797 non-null   object
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 893.4+ KB
```

[15]:
```
df.isna().sum()
```

[15]:
```
show_id            0
type               0
title              0
director        2624
cast             825
country          830
date_added         0
release_year       0
rating             7
duration           0
listed_in          0
description        0
dtype: int64
```

[16]:
```
df.head(5)
```

[16]:
```
  show_id     type                  title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                   cast        country  \
```

```
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…            NaN
3                                                NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                        description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[17]:
```
columns_to_impute = ['director', 'cast', 'country', 'rating']
df[columns_to_impute] = df[columns_to_impute].fillna("UNKNOWN")
```

[18]:
```
df.head(5)
```

[18]:
```
  show_id     type                  title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water          UNKNOWN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans          UNKNOWN
4      s5  TV Show           Kota Factory          UNKNOWN

                                                cast        country  \
0                                            UNKNOWN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…        UNKNOWN
3                                            UNKNOWN        UNKNOWN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…          India

  date_added  release_year rating   duration  \
```

```
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                     Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                         description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[19]:
```
df.isna().sum()
```

[19]:
```
show_id         0
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
description     0
dtype: int64
```

[20]:
```
df['director'] = df['director'].str.split(',')
```

[21]:
```
df['cast'] = df['cast'].str.split(',')
df.head(5)
```

[21]:
```
  show_id     type                  title           director  \
0      s1    Movie   Dick Johnson Is Dead  [Kirsten Johnson]
1      s2  TV Show          Blood & Water          [UNKNOWN]
2      s3  TV Show              Ganglands  [Julien Leclercq]
3      s4  TV Show  Jailbirds New Orleans          [UNKNOWN]
4      s5  TV Show           Kota Factory          [UNKNOWN]
```

```
                                                  cast        country  \
0                                            [UNKNOWN]  United States
1  [Ama Qamata,  Khosi Ngema,  Gail Mabalane,  Th…   South Africa
2  [Sami Bouajila,  Tracy Gotoas,  Samuel Jouy,  …        UNKNOWN
3                                            [UNKNOWN]        UNKNOWN
4  [Mayur More,  Jitendra Kumar,  Ranjan Raj,  Al…          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                            listed_in  \
0                                       Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act…
3                              Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV …

                                          description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[22]:
```
df['country'] = df['country'].str.split(',')
df['listed_in'] = df['listed_in'].str.split(',')
```

[23]:
```
df.head(5)
```

[23]:
```
  show_id     type                  title           director  \
0      s1    Movie   Dick Johnson Is Dead  [Kirsten Johnson]
1      s2  TV Show          Blood & Water          [UNKNOWN]
2      s3  TV Show              Ganglands  [Julien Leclercq]
3      s4  TV Show  Jailbirds New Orleans          [UNKNOWN]
4      s5  TV Show           Kota Factory          [UNKNOWN]

                                                  cast          country  \
0                                            [UNKNOWN]  [United States]
1  [Ama Qamata,  Khosi Ngema,  Gail Mabalane,  Th…   [South Africa]
2  [Sami Bouajila,  Tracy Gotoas,  Samuel Jouy,  …        [UNKNOWN]
3                                            [UNKNOWN]        [UNKNOWN]
4  [Mayur More,  Jitendra Kumar,  Ranjan Raj,  Al…          [India]
```

```
  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                   [Documentaries]
1  [International TV Shows,  TV Dramas,  TV Myste…
2  [Crime TV Shows,  International TV Shows,  TV …
3                          [Docuseries,  Reality TV]
4  [International TV Shows,  Romantic TV Shows,  …

                                        description
0  As her father nears the end of his life, filmm…
1  After crossing paths at a party, a Cape Town t…
2  To protect his family from a powerful drug lor…
3  Feuds, flirtations and toilet talk go down amo…
4  In a city of coaching centers known to train I…
```

[24]:
```
df = df.explode('director').explode('cast').explode('country').
 ↪explode('listed_in')
df.head(5)
```

[24]:
```
  show_id     type                 title         director          cast  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson       UNKNOWN
1      s2  TV Show         Blood & Water          UNKNOWN    Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN    Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN    Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Khosi Ngema

         country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons

                listed_in                                        description
0           Documentaries  As her father nears the end of his life, filmm…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
1               TV Dramas  After crossing paths at a party, a Cape Town t…
1            TV Mysteries  After crossing paths at a party, a Cape Town t…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
```

[25]:
```
df['cast'] = df['cast'].str.strip()
df['country'] = df['country'].str.strip()
df['director'] = df['director'].str.strip()
df['listed_in'] = df['listed_in'].str.strip()
df.head(5)
```

[25]:
```
  show_id     type                title         director         cast  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson      UNKNOWN
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN  Khosi Ngema

         country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons

                listed_in                                        description
0           Documentaries  As her father nears the end of his life, filmm…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
1               TV Dramas  After crossing paths at a party, a Cape Town t…
1            TV Mysteries  After crossing paths at a party, a Cape Town t…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
```

[26]:
```
df['duration']
```

[26]:
```
0          90 min
1       2 Seasons
1       2 Seasons
1       2 Seasons
1       2 Seasons
          …
8806      111 min
8806      111 min
8806      111 min
8806      111 min
8806      111 min
Name: duration, Length: 201907, dtype: object
```

[27]:
```
df[['duration_value', 'duration_unit']] = df['duration'].str.
 ↪extract(r'(\d+)\s*(\D+)')
df.head(5)
```

[27]:
```
  show_id     type                 title         director         cast  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson      UNKNOWN
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN  Khosi Ngema

         country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons

                listed_in                                        description  \
0           Documentaries  As her father nears the end of his life, filmm…
1  International TV Shows  After crossing paths at a party, a Cape Town t…
1               TV Dramas  After crossing paths at a party, a Cape Town t…
1            TV Mysteries  After crossing paths at a party, a Cape Town t…
1  International TV Shows  After crossing paths at a party, a Cape Town t…

  duration_value duration_unit
0             90           min
1              2       Seasons
1              2       Seasons
1              2       Seasons
1              2       Seasons
```

[28]:
```python
df = df.drop(columns=['duration'])
```

[29]:
```python
df.head(5)
```

[29]:
```
  show_id     type                 title         director         cast  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson      UNKNOWN
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN   Ama Qamata
1      s2  TV Show         Blood & Water          UNKNOWN  Khosi Ngema

         country date_added  release_year rating               listed_in  \
0  United States 2021-09-25          2020  PG-13           Documentaries
1   South Africa 2021-09-24          2021  TV-MA  International TV Shows
1   South Africa 2021-09-24          2021  TV-MA               TV Dramas
1   South Africa 2021-09-24          2021  TV-MA            TV Mysteries
1   South Africa 2021-09-24          2021  TV-MA  International TV Shows

                                        description duration_value  \
```

```
0  As her father nears the end of his life, filmm…            90
1  After crossing paths at a party, a Cape Town t…             2
1  After crossing paths at a party, a Cape Town t…             2
1  After crossing paths at a party, a Cape Town t…             2
1  After crossing paths at a party, a Cape Town t…             2

  duration_unit
0           min
1       Seasons
1       Seasons
1       Seasons
1       Seasons
```

[30]:
```
df.shape
```

[30]:
```
(201907, 13)
```

[31]:
```
df.info()
```

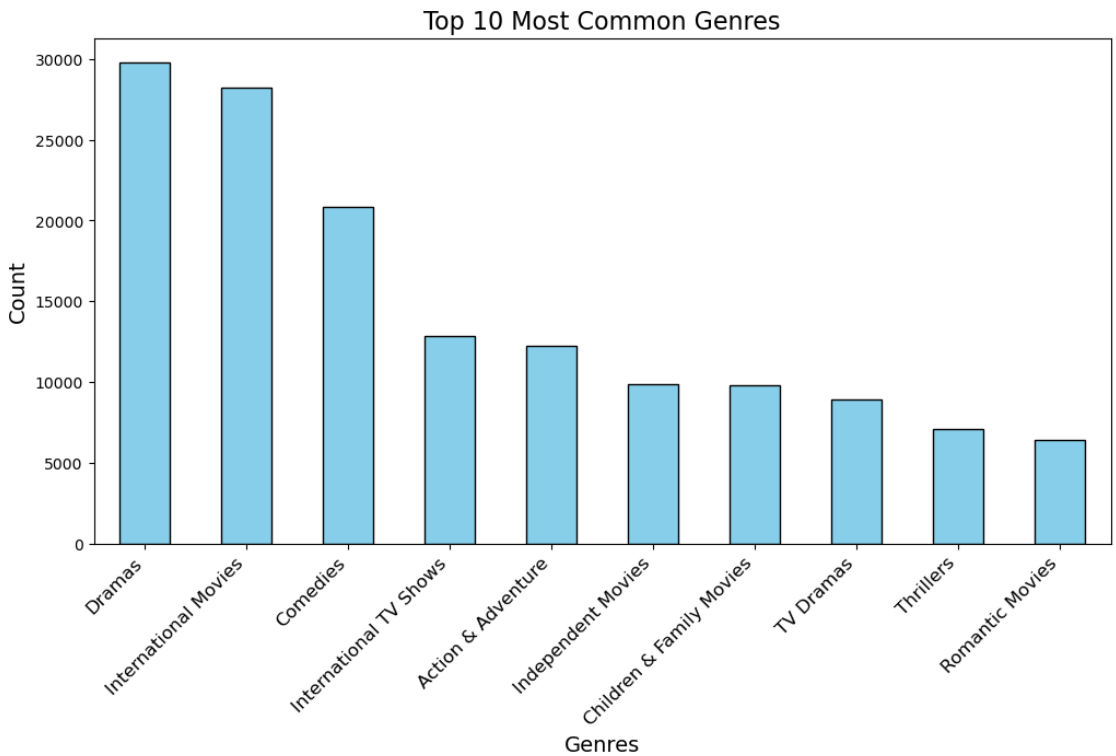
```
<class 'pandas.core.frame.DataFrame'>
Index: 201907 entries, 0 to 8806
Data columns (total 13 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   show_id         201907 non-null  object
 1   type            201907 non-null  object
 2   title           201907 non-null  object
 3   director        201907 non-null  object
 4   cast            201907 non-null  object
 5   country         201907 non-null  object
 6   date_added      201907 non-null  datetime64[ns]
 7   release_year    201907 non-null  int64
 8   rating          201907 non-null  object
 9   listed_in       201907 non-null  object
 10  description     201907 non-null  object
 11  duration_value  201907 non-null  object
 12  duration_unit   201907 non-null  object
dtypes: datetime64[ns](1), int64(1), object(11)
memory usage: 21.6+ MB
```

[32]:
```
df['duration_value'] = pd.to_numeric(df['duration_value'])
df.info()
```

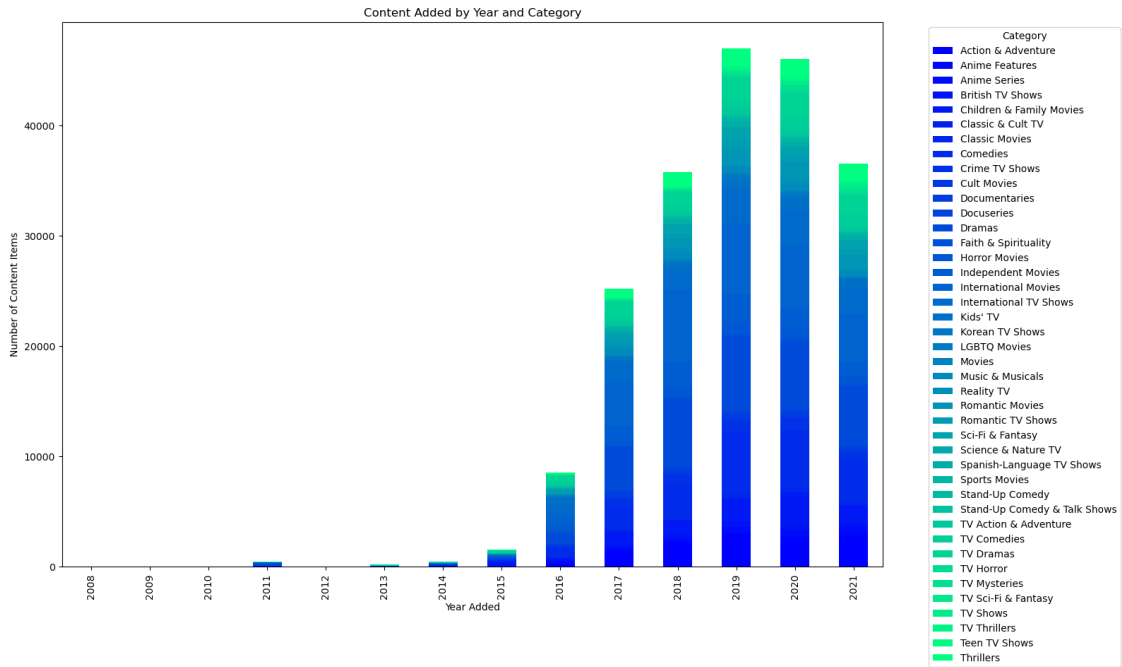
```
<class 'pandas.core.frame.DataFrame'>
Index: 201907 entries, 0 to 8806
Data columns (total 13 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
```

```
 0   show_id         201907 non-null  object
 1   type            201907 non-null  object
 2   title           201907 non-null  object
 3   director        201907 non-null  object
 4   cast            201907 non-null  object
 5   country         201907 non-null  object
 6   date_added      201907 non-null  datetime64[ns]
 7   release_year    201907 non-null  int64
 8   rating          201907 non-null  object
 9   listed_in       201907 non-null  object
 10  description     201907 non-null  object
 11  duration_value  201907 non-null  int64
 12  duration_unit   201907 non-null  object
dtypes: datetime64[ns](1), int64(2), object(10)
memory usage: 21.6+ MB
```

[33]:
```python
most_common_genre = df['listed_in'].value_counts()
most_common_genre.head(10)

import matplotlib.pyplot as plt
import seaborn as sns

top_10_genres = most_common_genre.head(10)

plt.figure(figsize=(12, 6))
top_10_genres.plot(kind='bar', color='skyblue', edgecolor='black')
plt.title('Top 10 Most Common Genres', fontsize=16)
plt.xlabel('Genres', fontsize=14)
plt.ylabel('Count', fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=12)

plt.show()
```



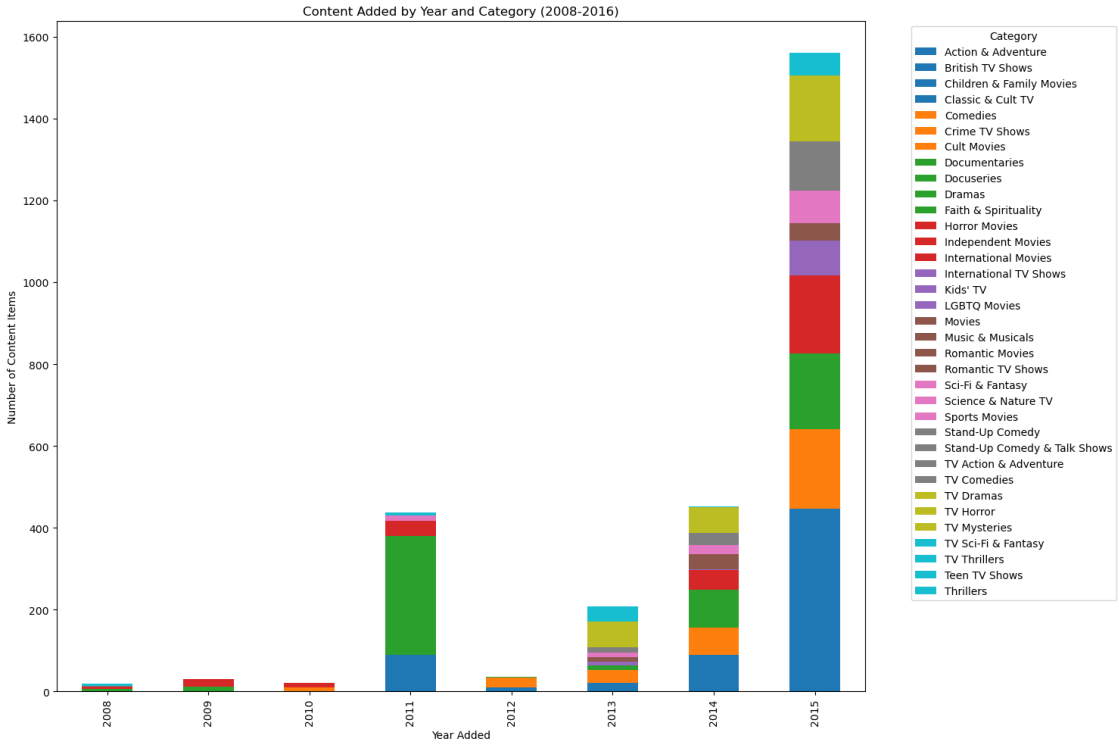
[34]:
```
df['year_added'] = df['date_added'].dt.year

content_added_by_year = df.groupby(['year_added', 'listed_in']).size().
 ↪reset_index(name='count')


pivot_table = content_added_by_year.pivot(index='year_added',␣
 ↪columns='listed_in', values='count').fillna(0)

pivot_table.plot(
    kind='bar',
    stacked=True,
    figsize=(15, 10),
    colormap='winter'
)

plt.title("Content Added by Year and Category")
plt.xlabel("Year Added")
plt.ylabel("Number of Content Items")
plt.legend(title="Category", bbox_to_anchor=(1.05, 1), loc='upper left')

plt.show()
```



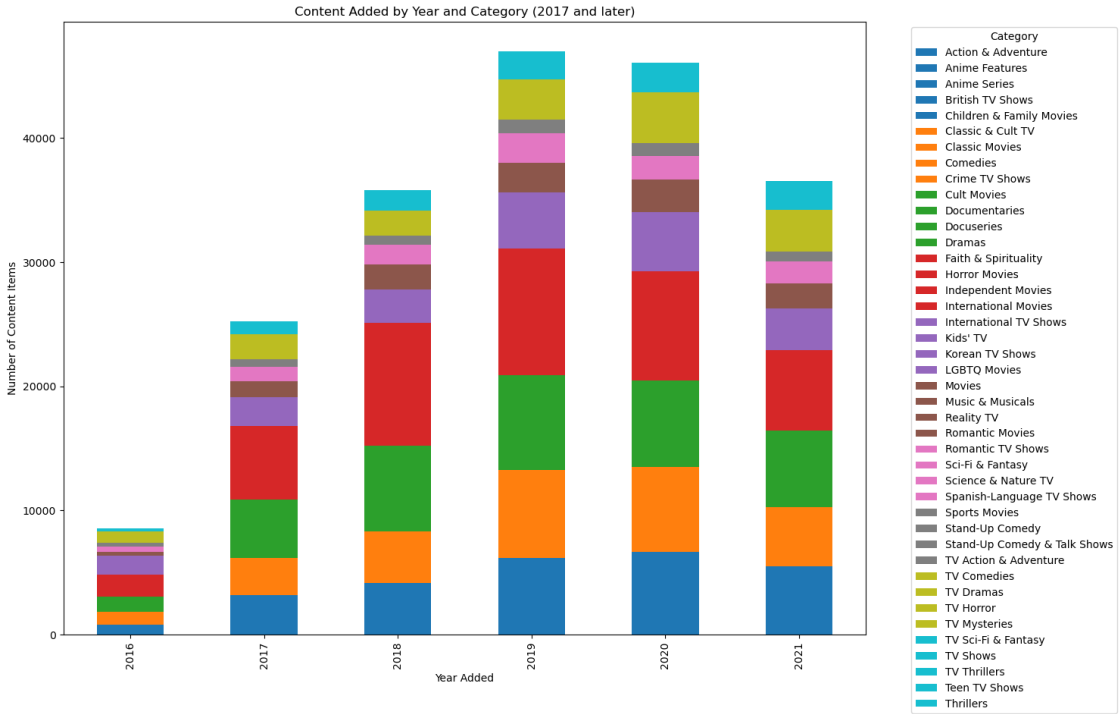
[35]:
```
subset_1 = content_added_by_year[content_added_by_year['year_added'] <= 2015]
subset_2 = content_added_by_year[content_added_by_year['year_added'] > 2015]


pivot_1 = subset_1.pivot(index='year_added', columns='listed_in',␣
 ↪values='count').fillna(0)
pivot_2 = subset_2.pivot(index='year_added', columns='listed_in',␣
 ↪values='count').fillna(0)

# Create the first graph (2008-2016)
plt.figure(figsize=(10, 6))
pivot_1.plot(kind='bar', stacked=True, figsize=(15, 10), colormap='tab10')
plt.title("Content Added by Year and Category (2008-2016)")
plt.xlabel("Year Added")
plt.ylabel("Number of Content Items")
plt.legend(title="Category", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

# Create the second graph (2017 onward)
plt.figure(figsize=(10, 6))
pivot_2.plot(kind='bar', stacked=True, figsize=(15, 10), colormap='tab10')
plt.title("Content Added by Year and Category (2017 and later)")
plt.xlabel("Year Added")
plt.ylabel("Number of Content Items")
```

```
plt.legend(title="Category", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

```
<Figure size 1000x600 with 0 Axes>
```




```
<Figure size 1000x600 with 0 Axes>
```



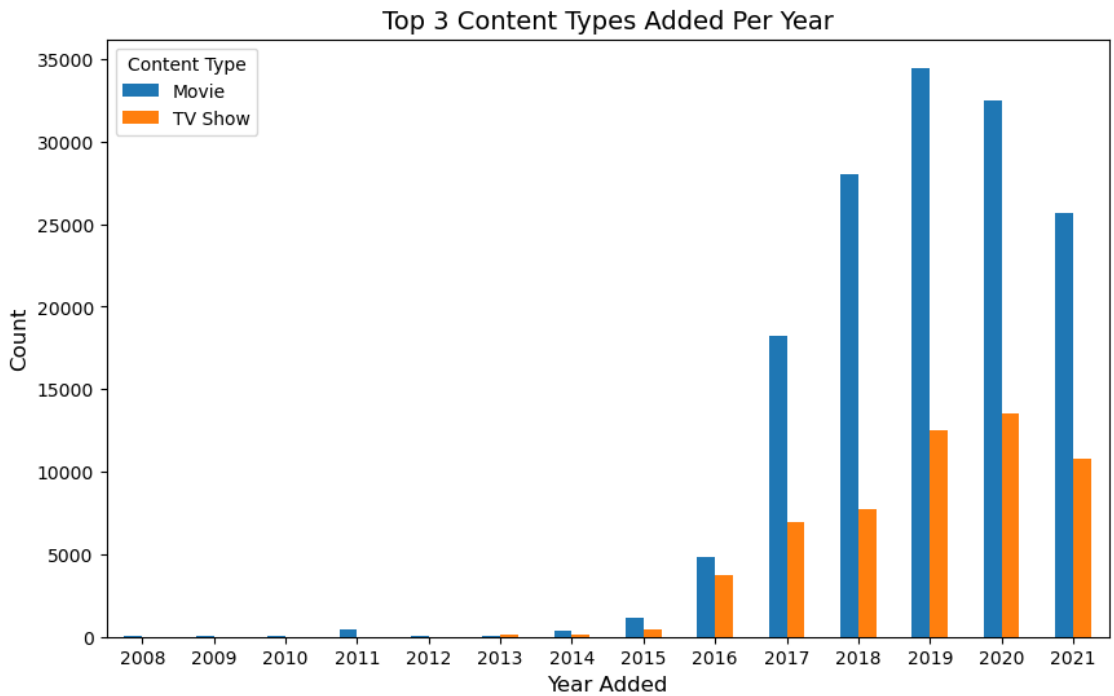
[36]:
```
top_content_per_year = df.groupby(['year_added', 'type']).size().
 ↪reset_index(name='count')
top_content_per_year_sorted = top_content_per_year.
 ↪sort_values(['year_added','type'], ascending=[True, True])
top_3_content = top_content_per_year_sorted.groupby('year_added').head(3)
top_3_content.head(5)
```

[36]:

	year_added	type	count
0	2008	Movie	18
1	2008	TV Show	1
2	2009	Movie	30
3	2010	Movie	20
4	2011	Movie	438

[37]:
```
pivot_data = top_3_content.pivot(index='year_added', columns='type',␣
 ↪values='count')

pivot_data.plot(kind='bar', figsize=(10, 6), stacked=False)

plt.title("Top 3 Content Types Added Per Year", fontsize=14)
plt.xlabel("Year Added", fontsize=12)
plt.ylabel("Count", fontsize=12)
plt.xticks(rotation=0)
plt.legend(title="Content Type", fontsize=10)
```

```
plt.show()
```



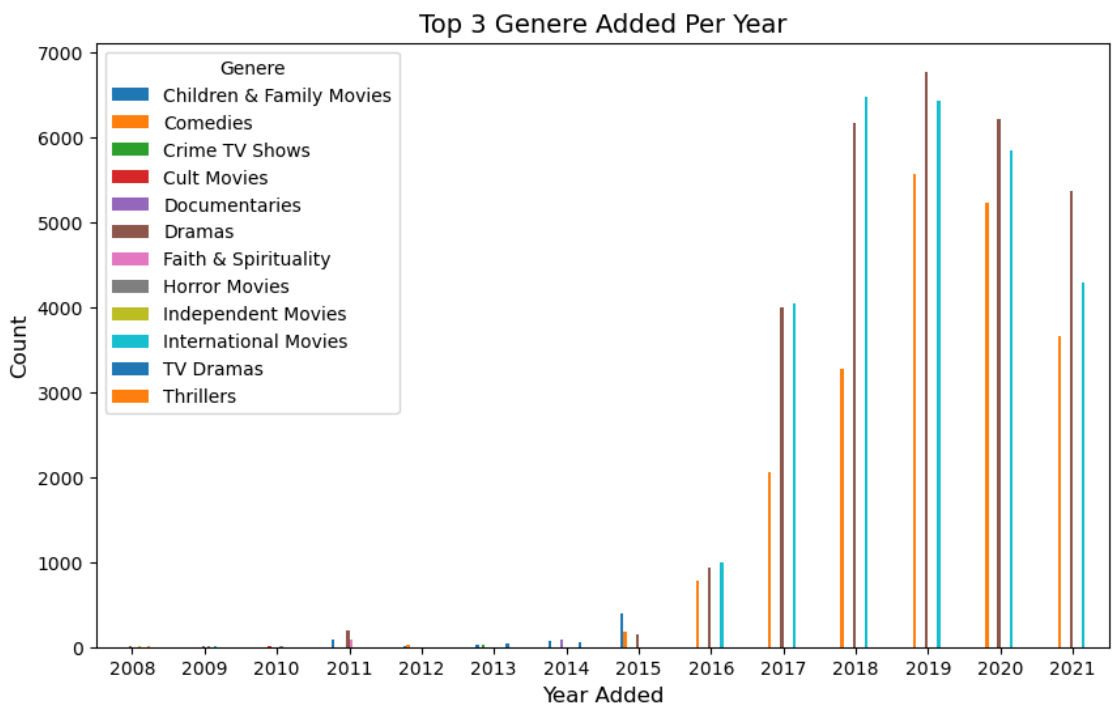
[38]:
```
top_genere_per_year = df.groupby(['year_added', 'listed_in']).size().
↪reset_index(name='count')
top_genere_per_year_sorted = top_genere_per_year.sort_values(['year_added',␣
↪'count'], ascending=[True, False])
top_3_genere = top_genere_per_year_sorted.groupby('year_added').head(3)
top_3_genere.head(5)
```

[38]:

	year_added	listed_in	count
0	2008	Dramas	6
1	2008	Independent Movies	6
3	2008	Thrillers	6
4	2009	Dramas	12
6	2009	International Movies	12

[39]:
```
pivot_data = top_3_genere.pivot(index='year_added', columns='listed_in',␣
↪values='count')

pivot_data.plot(kind='bar', figsize=(10, 6), stacked=False)

plt.title("Top 3 Genere Added Per Year", fontsize=14)
plt.xlabel("Year Added", fontsize=12)
```

```
plt.ylabel("Count", fontsize=12)
plt.xticks(rotation=0)
plt.legend(title="Genere", fontsize=10)

plt.show()
```



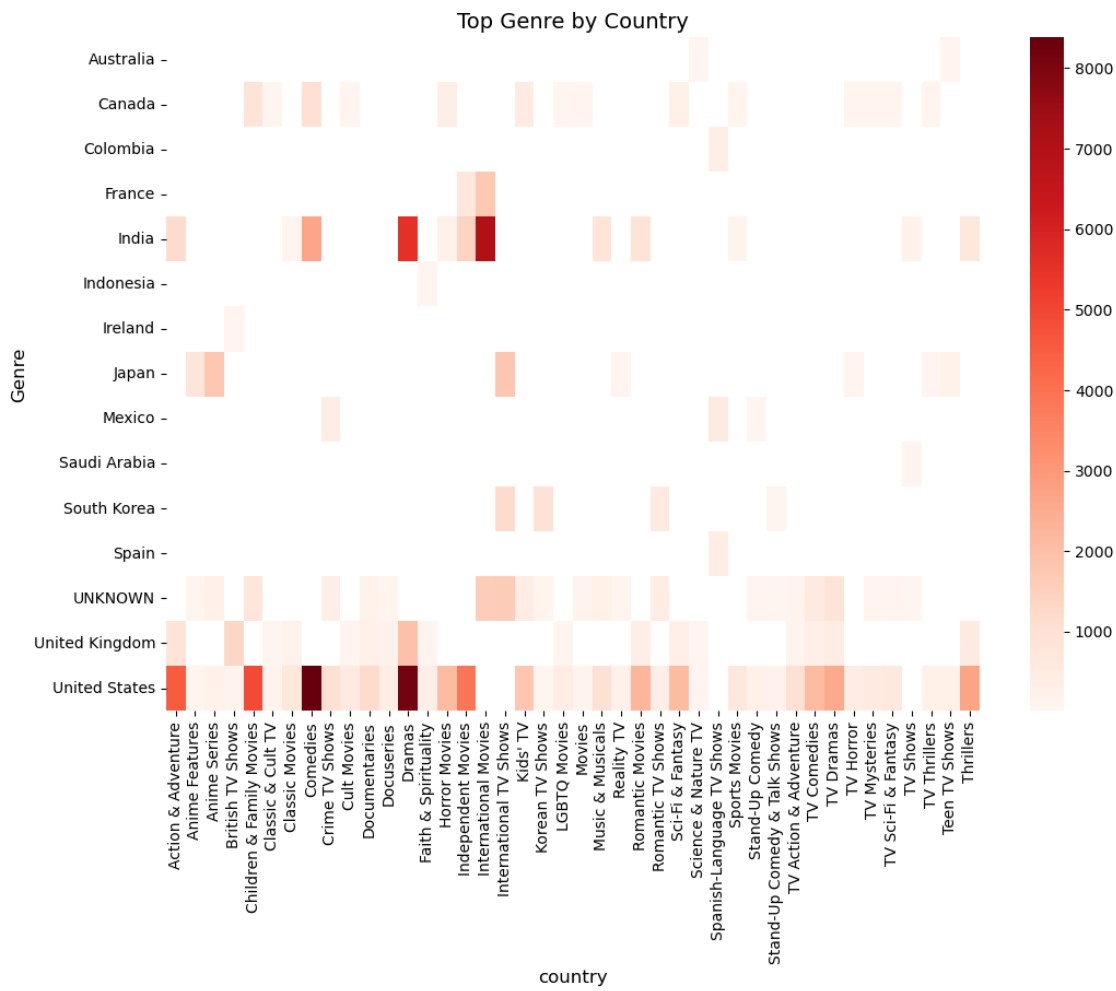
[41]:
```
top_genere_per_country = df.groupby(['listed_in', 'country']).size().
 ↪reset_index(name='count')
top_genere_per_country_sorted = top_genere_per_country.
 ↪sort_values(['listed_in', 'count'], ascending=[False, False])
top_genere_per_country = top_genere_per_country_sorted.groupby('listed_in').
 ↪head(3)
top_genere_per_country.head(5)
```

[41]:

	listed_in	country	count
1453	Thrillers	United States	2698
1419	Thrillers	India	743
1452	Thrillers	United Kingdom	541
1400	Teen TV Shows	United States	297
1393	Teen TV Shows	Japan	193

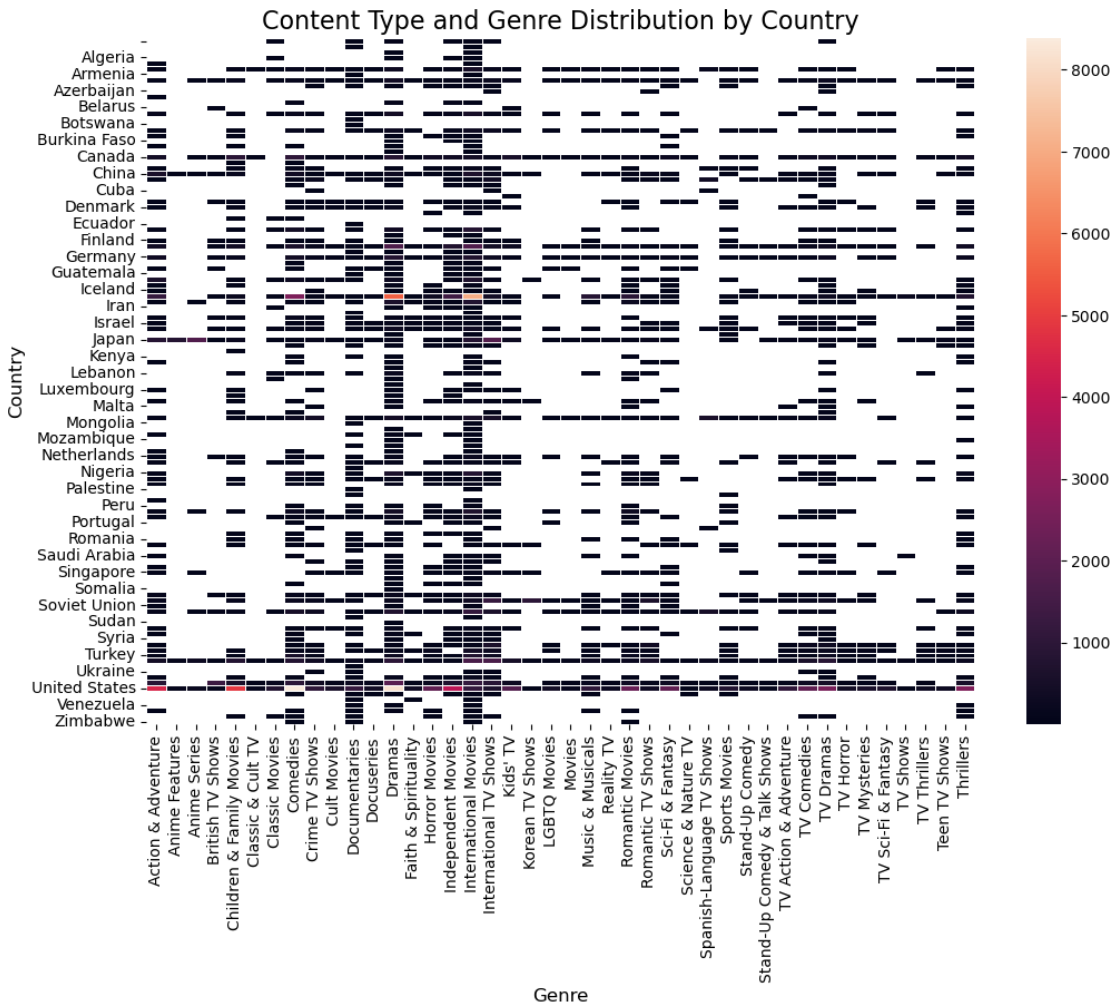
[42]:
```
heatmap_data = top_genere_per_country.pivot(index='country',␣
 ↪columns='listed_in', values='count')
```

```
plt.figure(figsize=(12, 8))
sns.heatmap(heatmap_data, cmap='Reds')

plt.title("Top Genre by Country", fontsize=14)
plt.xlabel("country", fontsize=12)
plt.ylabel("Genre", fontsize=12)
plt.xticks(rotation=90, ha='center')

plt.show()
```



[43]:
```
country_type_genre = df.groupby(["country", "type", "listed_in"]).size().
↪reset_index(name="count")

heatmap_data = country_type_genre.pivot_table(index="country",␣
↪columns="listed_in", values="count", aggfunc="sum")
```

```
plt.figure(figsize=(12, 8))
sns.heatmap(heatmap_data, linewidths=0.5)

plt.title("Content Type and Genre Distribution by Country", fontsize=16)
plt.xlabel("Genre", fontsize=12)
plt.ylabel("Country", fontsize=12)

plt.show()
```



[44]:
```
movies = df[df["type"] == "Movie"]
movies.head(5)
```

[44]:
```
  show_id   type                                title         director  \
0      s1  Movie                  Dick Johnson Is Dead  Kirsten Johnson
6      s7  Movie  My Little Pony: A New Generation    Robert Cullen
6      s7  Movie  My Little Pony: A New Generation    Robert Cullen
```

```
6      s7  Movie  My Little Pony: A New Generation    Robert Cullen
6      s7  Movie  My Little Pony: A New Generation    Robert Cullen

              cast        country date_added  release_year rating  \
0          UNKNOWN  United States 2021-09-25          2020  PG-13
6  Vanessa Hudgens        UNKNOWN 2021-09-24          2021     PG
6     Kimiko Glenn        UNKNOWN 2021-09-24          2021     PG
6    James Marsden        UNKNOWN 2021-09-24          2021     PG
6     Sofia Carson        UNKNOWN 2021-09-24          2021     PG

                 listed_in  \
0             Documentaries
6  Children & Family Movies
6  Children & Family Movies
6  Children & Family Movies
6  Children & Family Movies

                                        description  duration_value  \
0  As her father nears the end of his life, filmm…              90
6  Equestria's divided. But a bright-eyed hero be…              91
6  Equestria's divided. But a bright-eyed hero be…              91
6  Equestria's divided. But a bright-eyed hero be…              91
6  Equestria's divided. But a bright-eyed hero be…              91

  duration_unit  year_added
0           min        2021
6           min        2021
6           min        2021
6           min        2021
6           min        2021
```

[45]:
```
tv_shows = df[df["type"] == "TV Show"]
tv_shows.head(5)
```

[45]:
```
  show_id     type          title director         cast       country  \
1      s2  TV Show  Blood & Water  UNKNOWN   Ama Qamata  South Africa
1      s2  TV Show  Blood & Water  UNKNOWN   Ama Qamata  South Africa
1      s2  TV Show  Blood & Water  UNKNOWN   Ama Qamata  South Africa
1      s2  TV Show  Blood & Water  UNKNOWN  Khosi Ngema  South Africa
1      s2  TV Show  Blood & Water  UNKNOWN  Khosi Ngema  South Africa

  date_added  release_year rating               listed_in  \
1 2021-09-24          2021  TV-MA  International TV Shows
1 2021-09-24          2021  TV-MA               TV Dramas
1 2021-09-24          2021  TV-MA            TV Mysteries
1 2021-09-24          2021  TV-MA  International TV Shows
1 2021-09-24          2021  TV-MA               TV Dramas
```

```
                                          description  duration_value  \
1  After crossing paths at a party, a Cape Town t…               2
1  After crossing paths at a party, a Cape Town t…               2
1  After crossing paths at a party, a Cape Town t…               2
1  After crossing paths at a party, a Cape Town t…               2
1  After crossing paths at a party, a Cape Town t…               2

  duration_unit  year_added
1       Seasons        2021
1       Seasons        2021
1       Seasons        2021
1       Seasons        2021
1       Seasons        2021
```

[46]:
```
top_movie_directors = movies["director"].value_counts()
top_movie_directors.head(10)
```

[46]:
```
director
UNKNOWN                1285
Martin Scorsese         419
Youssef Chahine         409
Cathy Garcia-Molina     356
Steven Spielberg        355
Lars von Trier          336
Raja Gosnell            308
Tom Hooper              306
McG                     293
David Dhawan            270
Name: count, dtype: int64
```

[47]:
```
top_movie_cast = movies["cast"].value_counts()
top_movie_cast.head(10)
```

[47]:
```
cast
UNKNOWN             1331
Liam Neeson          161
Alfred Molina        157
John Krasinski       138
Salma Hayek          130
Frank Langella       128
Anupam Kher          118
John Rhys-Davies     116
Shah Rukh Khan       108
Naseeruddin Shah     106
Name: count, dtype: int64
```

[48]:
```
top_tv_directors = tv_shows["director"].value_counts()

top_tv_directors.head(10)
```

[48]:
```
director
UNKNOWN              49200
Noam Murro             189
Thomas Astruc          160
Houda Benyamina        104
Damien Chazelle        104
Alan Poul              104
Laïla Marrakchi        104
Rob Seidenglanz        103
Alejandro Lozano        90
Jay Oliva               81
Name: count, dtype: int64
```

[49]:
```
top_tv_cast = tv_shows["cast"].value_counts()
top_tv_cast.head(10)
```

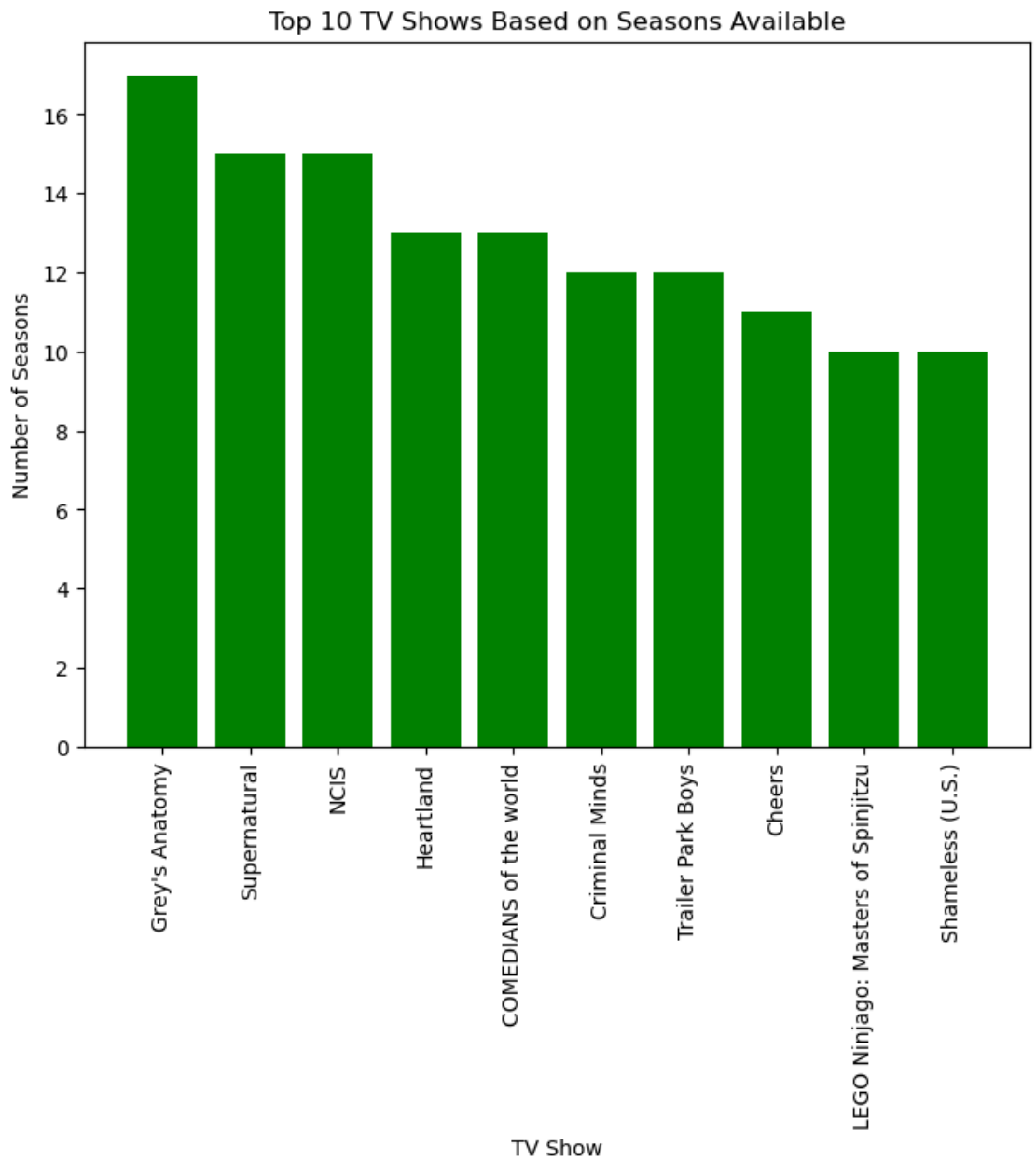
[49]:
```
cast
UNKNOWN                818
David Attenborough      82
Takahiro Sakurai        56
Yuki Kaji               45
Ai Kayano               41
Junichi Suwabe          39
Yuichi Nakamura         38
Daisuke Ono             38
Jun Fukuyama            38
Kate Harbour            37
Name: count, dtype: int64
```

[50]:
```
df["director_cast_pair"] = df["director"] + " & " + df["cast"]
top_director_cast_pairs = df["director_cast_pair"].value_counts()
top_director_cast_pairs.head(10)
```

[50]:
```
director_cast_pair
UNKNOWN & UNKNOWN                738
UNKNOWN & David Attenborough      72
UNKNOWN & Takahiro Sakurai        54
UNKNOWN & Yuki Kaji               43
UNKNOWN & Yuichi Nakamura         38
UNKNOWN & Jun Fukuyama            38
UNKNOWN & Kate Harbour            37
UNKNOWN & Junichi Suwabe          37
UNKNOWN & Ai Kayano               37
```

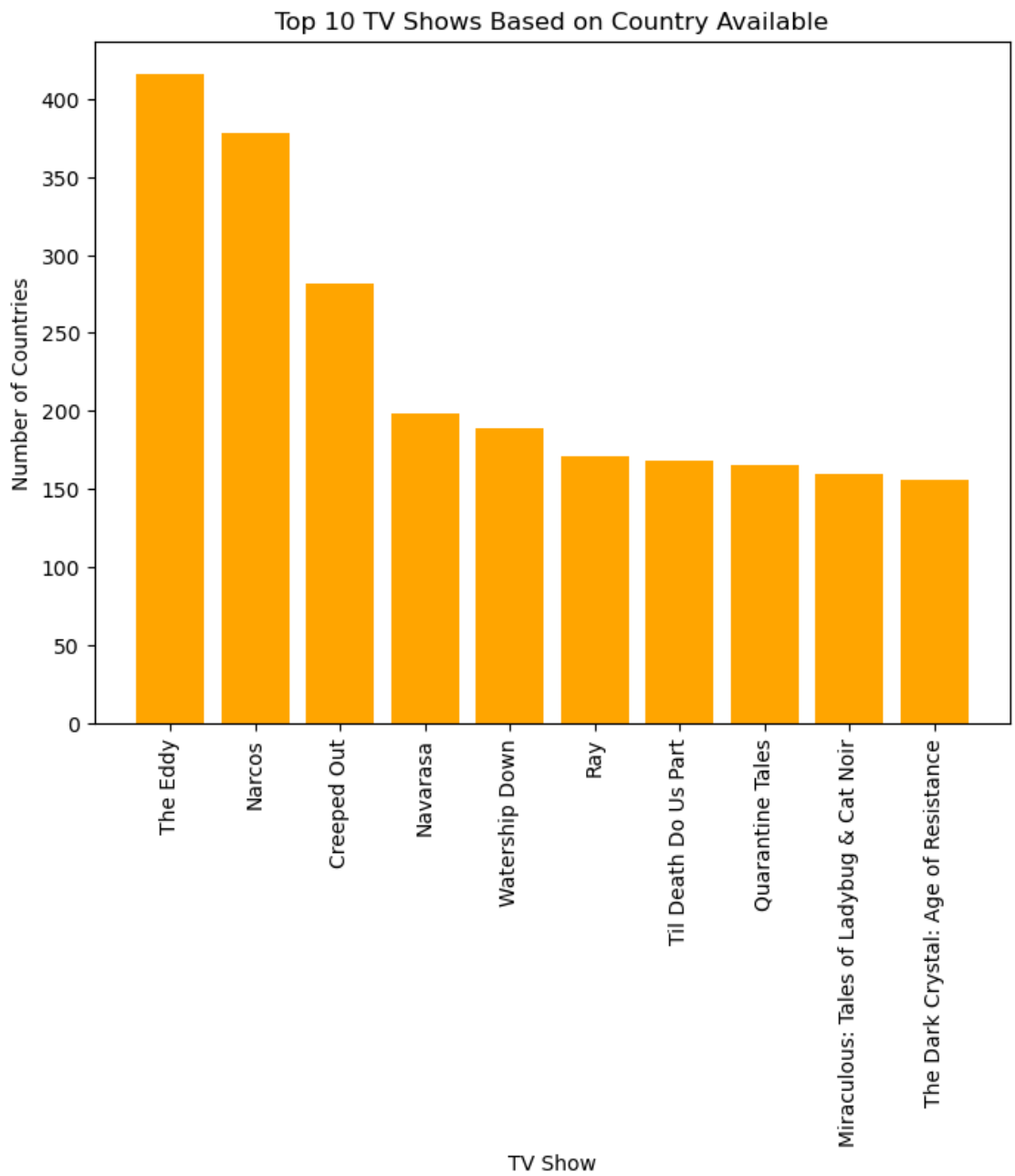
```
UNKNOWN & Daisuke Ono            36
Name: count, dtype: int64
```

[51]:
```
longest_shows = tv_shows.groupby(["show_id", "title"])["duration_value"].max().
 ↪reset_index(name="seasons").sort_values ('seasons', ascending=False).head(10)
plt.figure(figsize=(8, 6))
plt.bar(longest_shows["title"], longest_shows["seasons"], color="Green")
plt.title("Top 10 TV Shows Based on Seasons Available")
plt.xlabel("TV Show")
plt.ylabel("Number of Seasons")
plt.xticks(rotation=90)
plt.show()
```



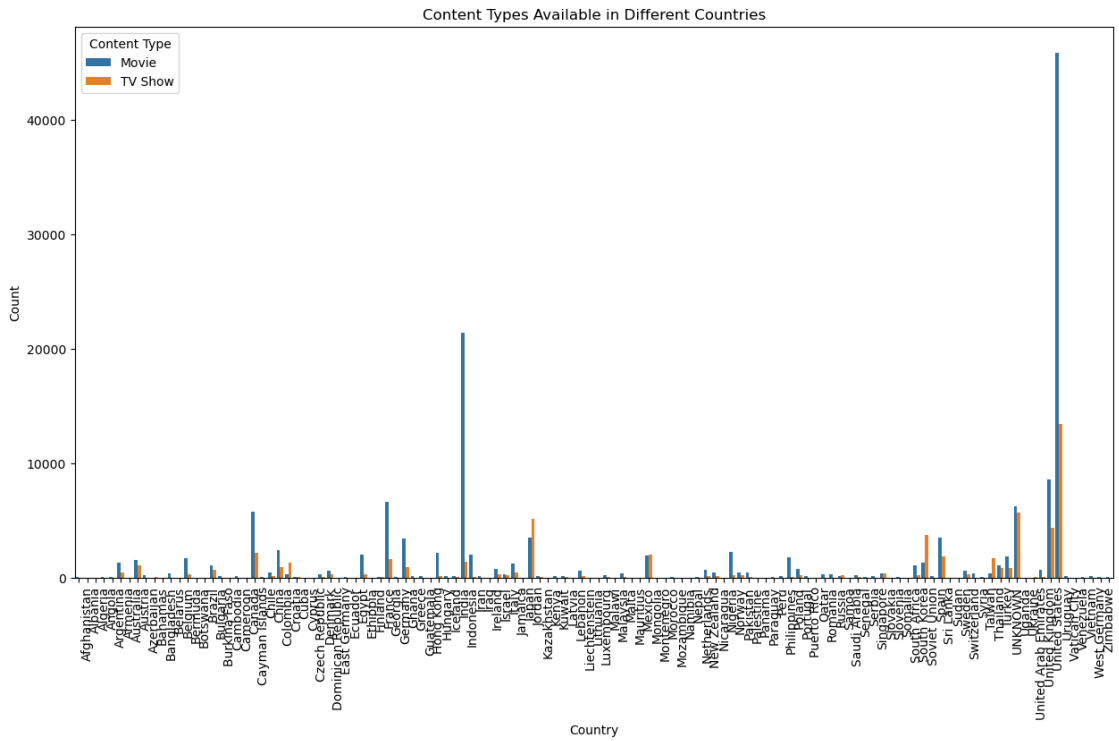
[52]:
```
widest_shows_geographically = tv_shows.groupby(["show_id", "title"])["country"].
↪count().reset_index(name="count").sort_values ('count', ascending=False).
↪head(10)
plt.figure(figsize=(8, 6))
plt.bar(widest_shows_geographically["title"],␣
↪widest_shows_geographically["count"], color="Orange")
plt.title("Top 10 TV Shows Based on Country Available")
plt.xlabel("TV Show")
plt.ylabel("Number of Countries")
```

```
plt.xticks(rotation=90)
plt.show()
```



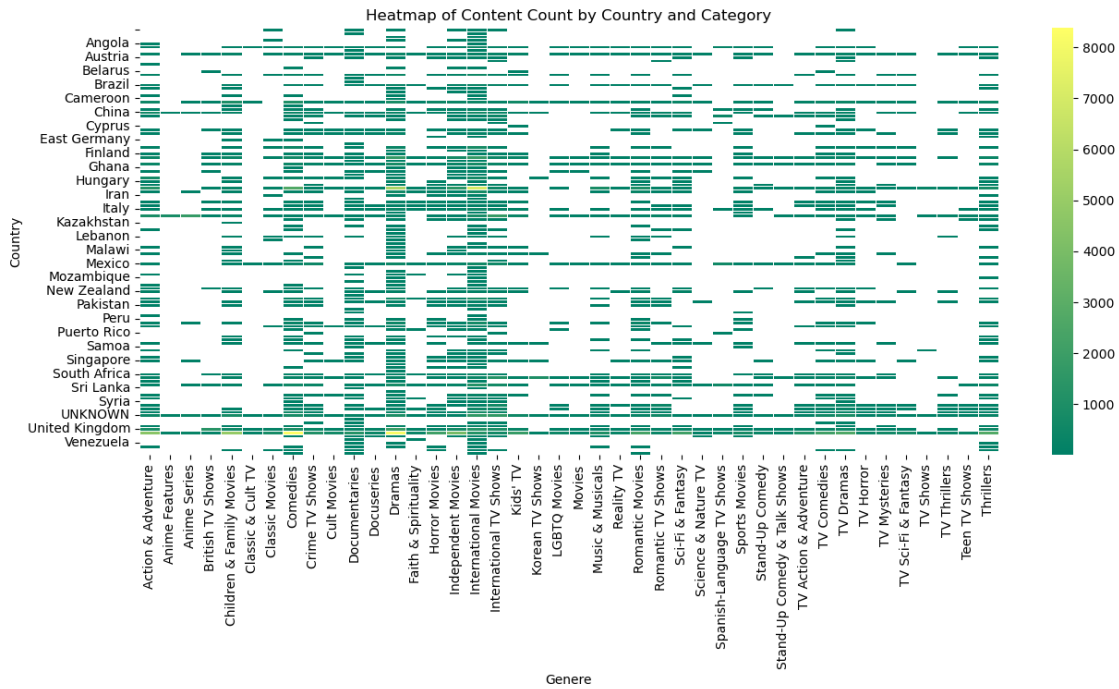
1 1. What type of content is available in different countries?

[53]:
```
content_by_country = df.groupby(['country', 'type'])['show_id'].count().
 ↪reset_index(name='count')

plt.figure(figsize=(15, 8))
sns.barplot(data=content_by_country, x='country', y='count', hue='type')
plt.title("Content Types Available in Different Countries")
plt.xticks(rotation=90)
plt.ylabel("Count")
plt.xlabel("Country")
plt.legend(title="Content Type")
plt.show()
```



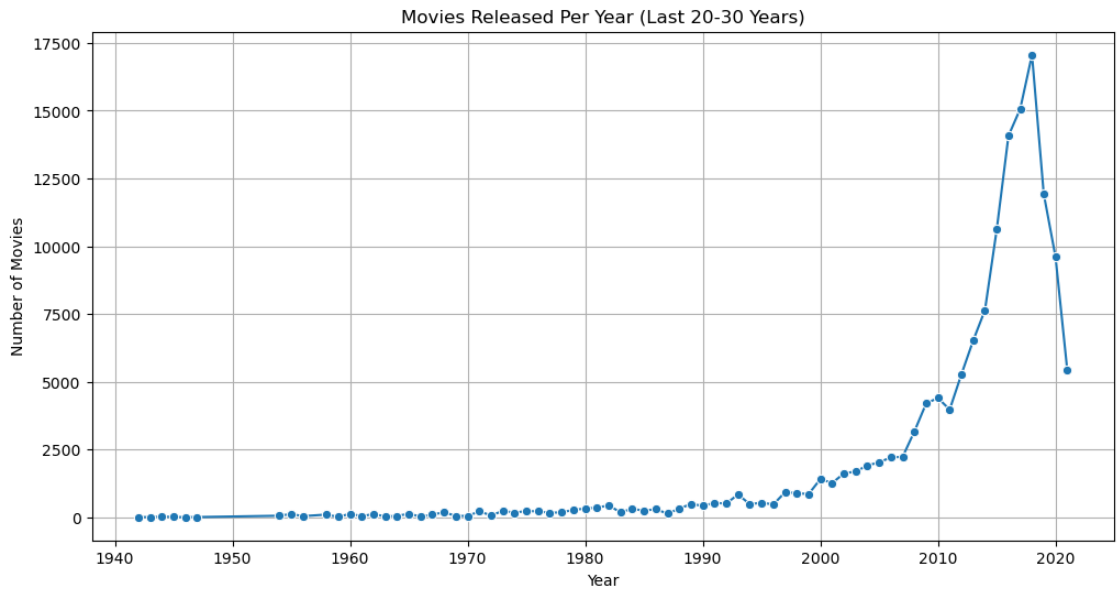
[54]:
```
content_by_country = df.groupby(['country', 'listed_in'])['show_id'].count().
 ↪reset_index(name='count')

heatmap_data = content_by_country.pivot(index='country', columns='listed_in',␣
 ↪values='count')

plt.figure(figsize=(15, 6))
sns.heatmap(heatmap_data,  cmap='summer', linewidths=0.5)
```

```
plt.title('Heatmap of Content Count by Country and Category')
plt.xlabel('Genere')
plt.ylabel('Country')

plt.show()
```



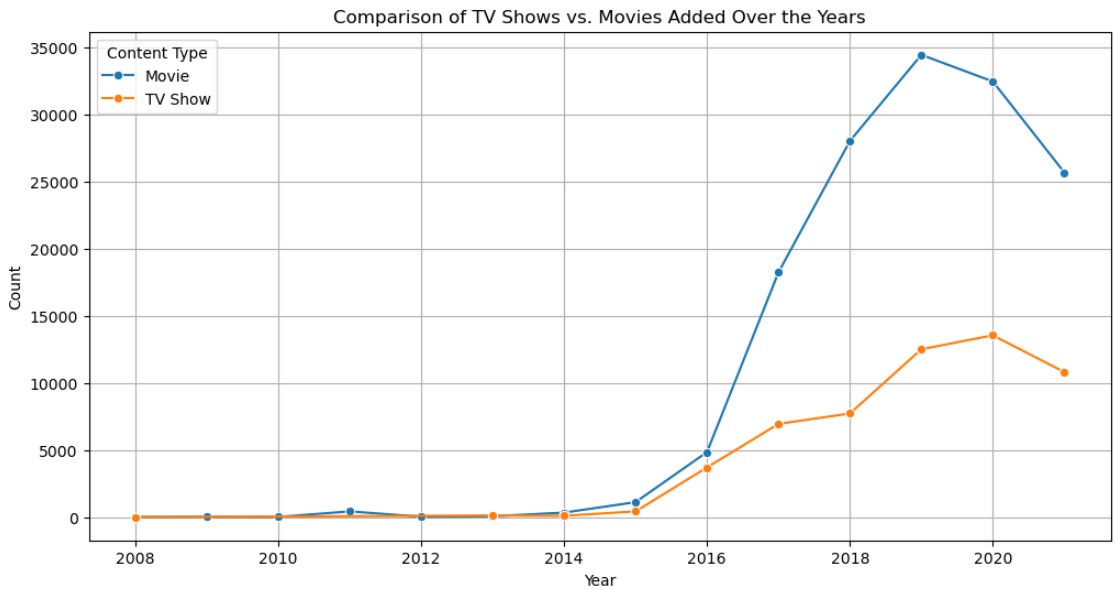
2 2. How has the number of movies released per year changed over the last 20-30 years?

[55]:
```
movies_per_year = df[df['type'] == 'Movie'].groupby('release_year')['show_id'].
↪count().reset_index(name='count')

plt.figure(figsize=(12, 6))
sns.lineplot(data=movies_per_year, x='release_year', y='count', marker='o')
plt.title("Movies Released Per Year (Last 20-30 Years)")
plt.xlabel("Year")
plt.ylabel("Number of Movies")
plt.grid(True)
plt.show()
```



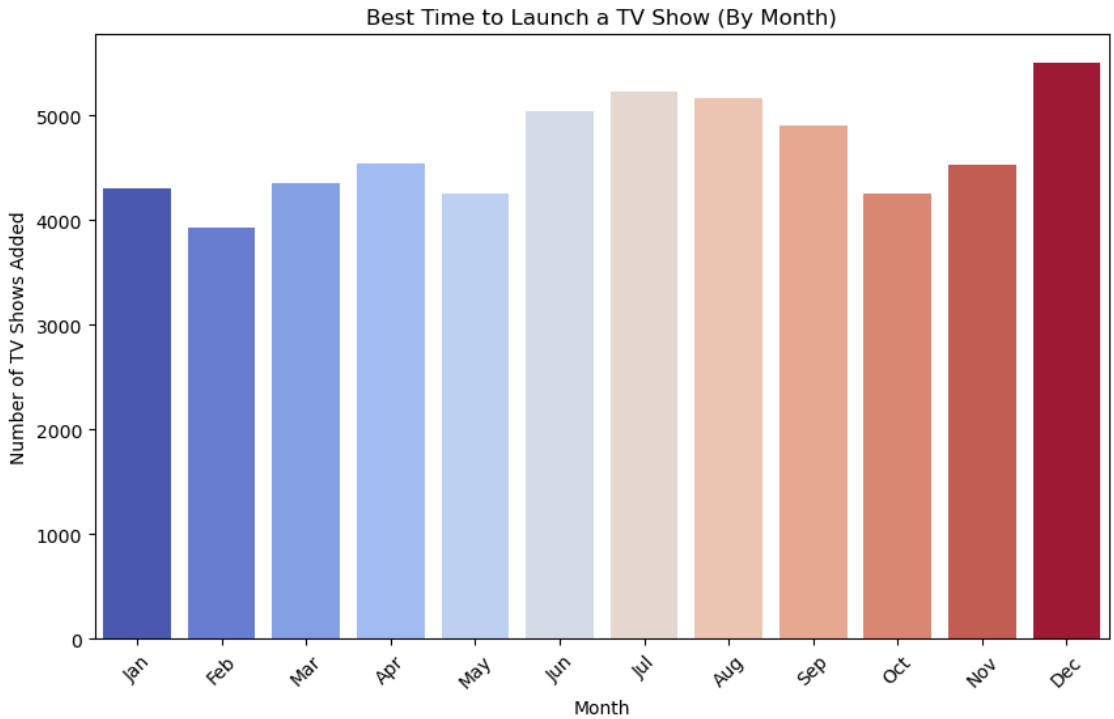
3 3. Comparison of TV Shows vs. Movies

[56]:
```
type_trend = df.groupby(['year_added', 'type'])['show_id'].count().
 ↪reset_index(name='count')

plt.figure(figsize=(12, 6))
sns.lineplot(data=type_trend, x='year_added', y='count', hue='type', marker='o')
plt.title("Comparison of TV Shows vs. Movies Added Over the Years")
plt.xlabel("Year")
plt.ylabel("Count")
plt.legend(title="Content Type")
plt.grid(True)
plt.show()
```



4 4. What is the best time to launch a TV show?

[57]:
```
df['month_added'] = pd.to_datetime(df['date_added']).dt.month

tv_shows = df[df['type'] == 'TV Show']

tv_shows_by_month = tv_shows['month_added'].value_counts().sort_index()

plt.figure(figsize=(10, 6))
sns.barplot(x=tv_shows_by_month.index, y=tv_shows_by_month.
 ↪values,hue=tv_shows_by_month.index, palette="coolwarm",legend=False)
plt.title("Best Time to Launch a TV Show (By Month)")
plt.xlabel("Month")
plt.ylabel("Number of TV Shows Added")
plt.xticks(range(12), ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug',␣
 ↪'Sep', 'Oct', 'Nov', 'Dec'], rotation=45)
plt.show()
```



5 5. Analysis of Actors/Directors of Different Types of Shows/Movies

[58]:
```
top_directors = df.groupby(['type', 'director'])['show_id'].count().
 ↪reset_index(name='count').sort_values(by=['type', 'count'], ascending=False)
top_cast = df.groupby(['type', 'cast'])['show_id'].count().
 ↪reset_index(name='count').sort_values(by=['type', 'count'], ascending=False)

print("Top Directors:\n", top_directors.head(10))
print("\nTop Cast:\n", top_cast.head(10))
```

```
Top Directors:
          type          director  count
5062  TV Show           UNKNOWN  49200
4975  TV Show        Noam Murro    189
5053  TV Show     Thomas Astruc    160
4785  TV Show         Alan Poul    104
4831  TV Show   Damien Chazelle    104
4877  TV Show   Houda Benyamina    104
4936  TV Show   Laïla Marrakchi    104
5013  TV Show   Rob Seidenglanz    103
4789  TV Show  Alejandro Lozano     90
4898  TV Show         Jay Oliva     81
```

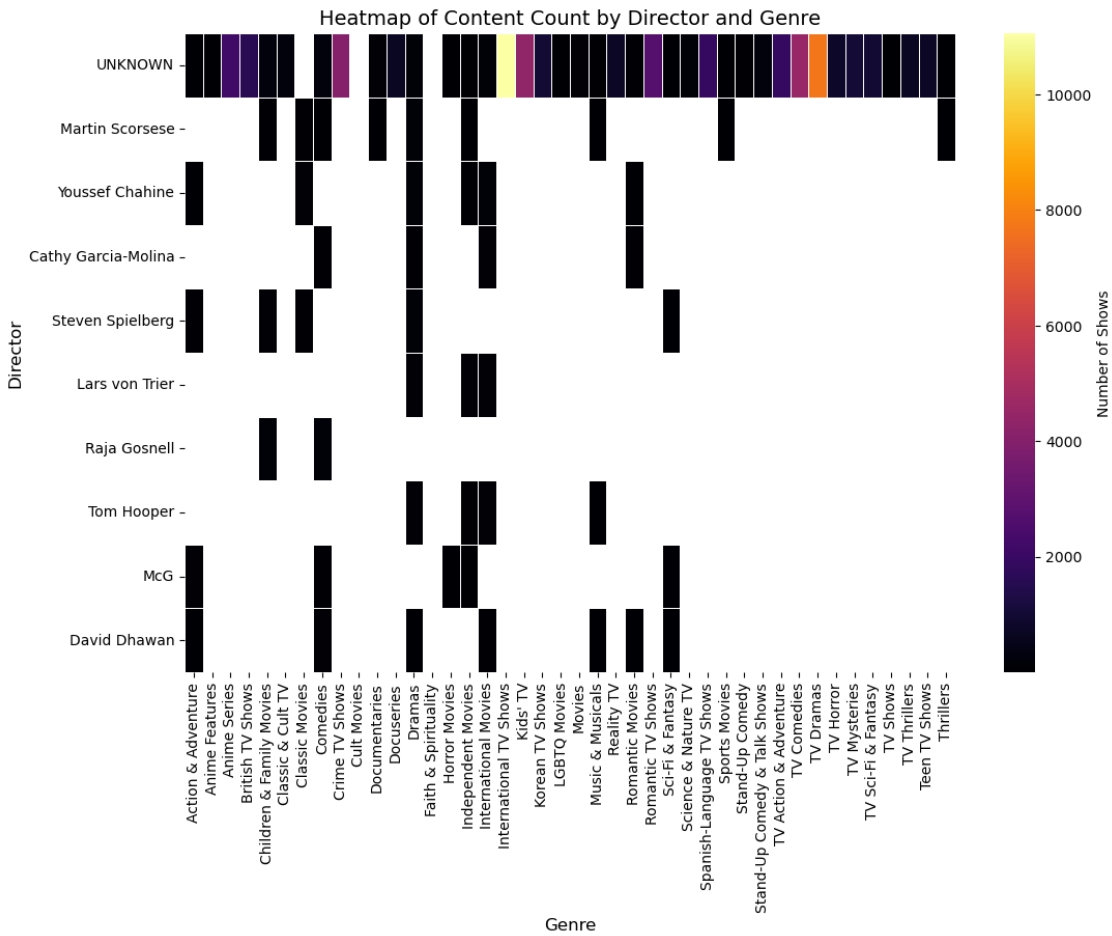
```
Top Cast:
           type                cast  count
39771  TV Show             UNKNOWN    818
29065  TV Show  David Attenborough     82
39137  TV Show    Takahiro Sakurai     56
40485  TV Show           Yuki Kaji     45
26203  TV Show           Ai Kayano     41
32731  TV Show      Junichi Suwabe     39
28815  TV Show         Daisuke Ono     38
32688  TV Show        Jun Fukuyama     38
40469  TV Show     Yuichi Nakamura     38
32967  TV Show        Kate Harbour     37
```

[59]:
```
top_directors_with_genres = (
    df.groupby(['type', 'director'])
    .agg(
        count=('show_id', 'count'),
        genres=('listed_in', lambda x: ', '.join(set(x)))
    )
    .reset_index()
    .sort_values(by=['type', 'count'], ascending=False)
)
top_directors_with_genres.head(10)
```

[59]:
```
         type          director  count  \
5062  TV Show           UNKNOWN  49200
4975  TV Show        Noam Murro    189
5053  TV Show     Thomas Astruc    160
4785  TV Show         Alan Poul    104
4831  TV Show   Damien Chazelle    104
4877  TV Show   Houda Benyamina    104
4936  TV Show   Laïla Marrakchi    104
5013  TV Show   Rob Seidenglanz    103
4789  TV Show  Alejandro Lozano     90
4898  TV Show         Jay Oliva     81

                                                 genres
5062  Crime TV Shows, Korean TV Shows, Science & Nat…
4975  International TV Shows, British TV Shows, TV A…
5053                    Kids' TV, TV Action & Adventure
4785                  International TV Shows, TV Dramas
4831                  International TV Shows, TV Dramas
4877                  International TV Shows, TV Dramas
4936                  International TV Shows, TV Dramas
5013  TV Sci-Fi & Fantasy, TV Mysteries, Crime TV Sh…
4789  International TV Shows, Crime TV Shows, Spanis…
```

```
4898            Crime TV Shows, TV Horror, Anime Series
```

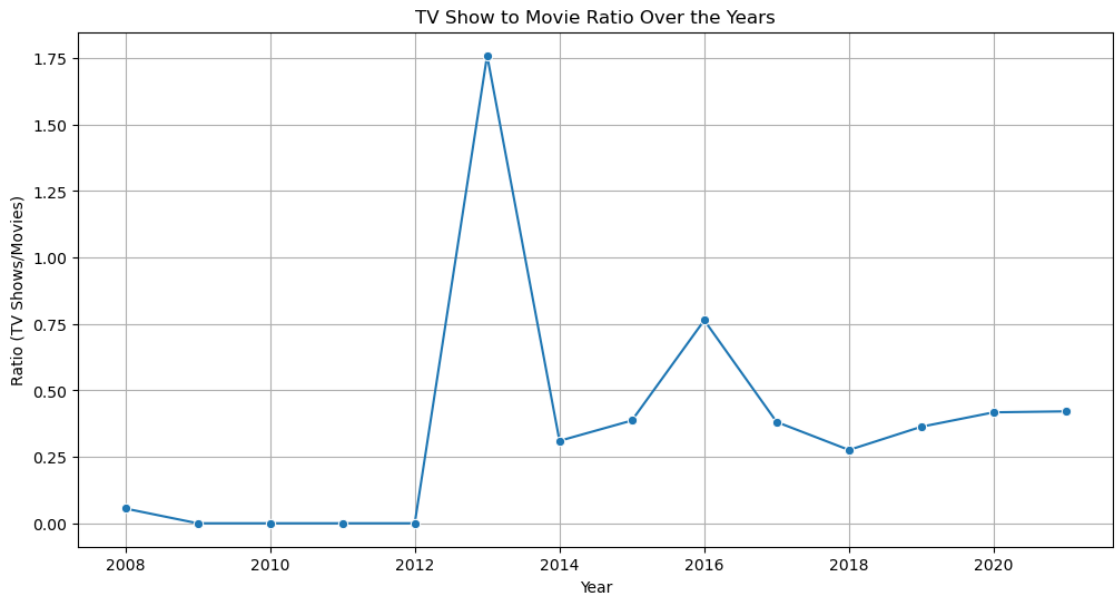
[60]:
```
heatmap_data = (
    df.groupby(['director', 'listed_in'])
    .size()
    .unstack()
)

top_directors = heatmap_data.sum(axis=1).nlargest(10).index
filtered_data = heatmap_data.loc[top_directors]

plt.figure(figsize=(12, 8))
sns.heatmap(
    filtered_data,
    cmap="inferno",
    linewidths=0.5,
    cbar_kws={'label': 'Number of Shows'}
)

plt.title("Heatmap of Content Count by Director and Genre", fontsize=14)
plt.xlabel("Genre", fontsize=12)
plt.ylabel("Director", fontsize=12)
plt.xticks(rotation=90, ha='center')
plt.yticks(rotation=0)
plt.show()
```



6 6. Does Netflix focus more on TV Shows than Movies in recent years?

[61]:
```
type_year_trend = df.groupby(['year_added', 'type'])['show_id'].count().
 ↪reset_index(name='count')

type_year_pivot = type_year_trend.pivot(index='year_added', columns='type',␣
 ↪values='count').fillna(0)
type_year_pivot['TV_to_Movie_Ratio'] = type_year_pivot['TV Show'] /␣
 ↪type_year_pivot['Movie']

plt.figure(figsize=(12, 6))
sns.lineplot(data=type_year_pivot, x=type_year_pivot.index,␣
 ↪y='TV_to_Movie_Ratio', marker='o')
plt.title("TV Show to Movie Ratio Over the Years")
plt.xlabel("Year")
```

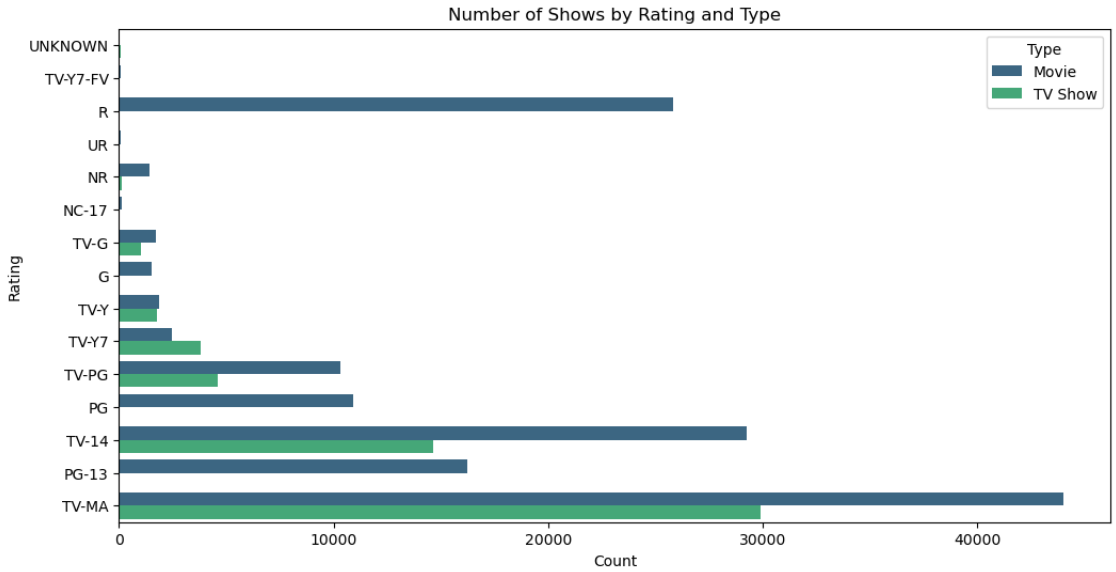
```
plt.ylabel("Ratio (TV Shows/Movies)")
plt.grid(True)
plt.show()
```



[62]:
```
df_grouped = df.groupby(['rating', 'type'])['show_id'].count().
↪reset_index(name='count').sort_values('count', ascending=True)
df_grouped.head(5)
```

[62]:

	rating	type	count
22	UNKNOWN	Movie	12
21	TV-Y7-FV	TV Show	24
7	R	TV Show	54
23	UNKNOWN	TV Show	58
20	TV-Y7-FV	Movie	62

[63]:
```
plt.figure(figsize=(12, 6))
sns.barplot(
    data=df_grouped,
    x='count',
    y='rating',
    hue='type',
    palette='viridis'
)

plt.title("Number of Shows by Rating and Type")
plt.xlabel("Count")
plt.ylabel("Rating")
```

```
plt.legend(title="Type", loc='upper right')

plt.show()
```